

NT-C-006 | AWS with DevOps | Course Knowledge Profile

Identity, purpose, prerequisites, scope and boundaries

Category	Cloud & DevOps
Target learners	Learners targeting cloud support, infrastructure automation and junior DevOps roles.
Prerequisites	Basic networking and operating-system knowledge; Linux foundation may be required.
Approved duration	Approved working range: 16-20 weeks / 180-240 guided hours.
Final outcome	Deploy and operate a secure cloud application using AWS services, containers, CI/CD, infrastructure as code and monitoring.
Assessment profile	Infrastructure/Security

Approved tools / platforms

AWS IAM/VPC/EC2/S3/RDS/CloudWatch, Git, Linux, Docker, CI/CD tooling, Terraform or CloudFormation; exact stack to confirm.

Knowledge boundaries

Include	Quality rule	Exclude / escalate
The eight approved modules, four sections, named tools, projects and mapped entry-level roles.	Require least privilege, authorised labs, configuration evidence, logs, monitoring, rollback and documented troubleshooting.	Never recommend unsafe production testing or treat a scanner output as a verified finding without validation.
Current product/platform procedures only when version and date are known.	Use primary documentation and record the source version.	Do not invent current commands, prices, tax rates, policies or certification status.
Evidence-based career guidance.	Cite tests, projects, capstone, viva and verified resume evidence.	No guaranteed job, placement, salary, interview or employer claim.

LLM answer pattern: identify the course and version, answer from approved records, cite record IDs, distinguish verified facts from guidance, and state any missing evidence.

NT-C-006 | AWS with DevOps | NT-C-006-S01 Deep Knowledge

Modules 1-2 | objectives, concepts, evidence and remediation

Section competency	Linux, networking, Git, delivery lifecycle; IAM, VPC, EC2, S3, RDS
Completion evidence	Version-controlled lab; Secure cloud environment
Section weight	20% of the weighted mock-test average
Assessment	10 MCQ (30 marks) + 4 troubleshooting scenarios (30) + 1 authorised practical lab (40)

Module 1 - DevOps foundation

Objective: Linux, networking, Git, delivery lifecycle

Observable evidence: Version-controlled lab

Concept	Approved knowledge statement	Application behaviour	Evidence
Linux	A defined competency within AWS with DevOps. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when linux is appropriate; apply it in a AWS with DevOps task; verify the result.	Version-controlled lab
networking	A defined competency within AWS with DevOps. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when networking is appropriate; apply it in a AWS with DevOps task; verify the result.	Version-controlled lab
Git	Distributed version control for tracking and reviewing source changes. In AWS with DevOps, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when git is appropriate; apply it in a AWS with DevOps task; verify the result.	Version-controlled lab
delivery lifecycle	A defined competency within AWS with DevOps. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when delivery lifecycle is appropriate; apply it in a AWS with DevOps task; verify the result.	Version-controlled lab

Module 2 - AWS core

Objective: IAM, VPC, EC2, S3, RDS

Observable evidence: Secure cloud environment

Concept	Approved knowledge statement	Application behaviour	Evidence
IAM	Identity and access management based on users, roles, policies and least privilege. In AWS with DevOps, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when iam is appropriate; apply it in a AWS with DevOps task; verify the result.	Secure cloud environment
VPC	An isolated logical cloud network. In AWS with DevOps, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when vpc is appropriate; apply it in a AWS with DevOps task; verify the result.	Secure cloud environment
EC2	A defined competency within AWS with DevOps. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when ec2 is appropriate; apply it in a AWS with DevOps task; verify the result.	Secure cloud environment
S3	A defined competency within AWS with DevOps. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when s3 is appropriate; apply it in a AWS with DevOps task; verify the result.	Secure cloud environment
RDS	A defined competency within AWS with DevOps. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when rds is appropriate; apply it in a AWS with DevOps task; verify the result.	Secure cloud environment

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Version-controlled lab; Secure cloud environment, return to Modules 1-2, complete one guided practice and then use a new equivalent test form.

NT-C-006 | AWS with DevOps | NT-C-006-S02 Deep Knowledge

Modules 3-4 | objectives, concepts, evidence and remediation

Section competency	Load balancing, Auto Scaling, DNS basics; Shell/Python basics, configuration tasks
Completion evidence	Highly available design; Automation scripts
Section weight	25% of the weighted mock-test average
Assessment	10 MCQ (30 marks) + 4 troubleshooting scenarios (30) + 1 authorised practical lab (40)

Module 3 - Availability

Objective: Load balancing, Auto Scaling, DNS basics

Observable evidence: Highly available design

Concept	Approved knowledge statement	Application behaviour	Evidence
Load balancing	A defined competency within AWS with DevOps. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when load balancing is appropriate; apply it in a AWS with DevOps task; verify the result.	Highly available design
Auto Scaling	A defined competency within AWS with DevOps. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when auto scaling is appropriate; apply it in a AWS with DevOps task; verify the result.	Highly available design
DNS basics	A defined competency within AWS with DevOps. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when dns basics is appropriate; apply it in a AWS with DevOps task; verify the result.	Highly available design

Module 4 - Automation

Objective: Shell/Python basics, configuration tasks

Observable evidence: Automation scripts

Concept	Approved knowledge statement	Application behaviour	Evidence
Shell/Python basics	A defined competency within AWS with DevOps. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when shell/python basics is appropriate; apply it in a AWS with DevOps task; verify the result.	Automation scripts
configuration tasks	A defined competency within AWS with DevOps. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when configuration tasks is appropriate; apply it in a AWS with DevOps task; verify the result.	Automation scripts

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Highly available design; Automation scripts, return to Modules 3-4, complete one guided practice and then use a new equivalent test form.

NT-C-006 | AWS with DevOps | NT-C-006-S03 Deep Knowledge

Modules 5-6 | objectives, concepts, evidence and remediation

Section competency	Docker, images, registries, ECR; Build, test, deploy pipelines; approvals
Completion evidence	Containerised service; Working pipeline
Section weight	25% of the weighted mock-test average
Assessment	10 MCQ (30 marks) + 4 troubleshooting scenarios (30) + 1 authorised practical lab (40)

Module 5 - Containers

Objective: Docker, images, registries, ECR

Observable evidence: Containerised service

Concept	Approved knowledge statement	Application behaviour	Evidence
Docker	A defined competency within AWS with DevOps. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when docker is appropriate; apply it in a AWS with DevOps task; verify the result.	Containerised service
images	A defined competency within AWS with DevOps. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when images is appropriate; apply it in a AWS with DevOps task; verify the result.	Containerised service
registries	A defined competency within AWS with DevOps. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when registries is appropriate; apply it in a AWS with DevOps task; verify the result.	Containerised service
ECR	A defined competency within AWS with DevOps. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when ecr is appropriate; apply it in a AWS with DevOps task; verify the result.	Containerised service

Module 6 - CI/CD

Objective: Build, test, deploy pipelines; approvals

Observable evidence: Working pipeline

Concept	Approved knowledge statement	Application behaviour	Evidence
Build	A defined competency within AWS with DevOps. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when build is appropriate; apply it in a AWS with DevOps task; verify the result.	Working pipeline
test	A defined competency within AWS with DevOps. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when test is appropriate; apply it in a AWS with DevOps task; verify the result.	Working pipeline
deploy pipelines	A defined competency within AWS with DevOps. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when deploy pipelines is appropriate; apply it in a AWS with DevOps task; verify the result.	Working pipeline
approvals	A defined competency within AWS with DevOps. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when approvals is appropriate; apply it in a AWS with DevOps task; verify the result.	Working pipeline

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Containerised service; Working pipeline, return to Modules 5-6, complete one guided practice and then use a new equivalent test form.

NT-C-006 | AWS with DevOps | NT-C-006-S04 Deep Knowledge

Modules 7-8 | objectives, concepts, evidence and remediation

Section competency	Terraform/CloudFormation, state, review; CloudWatch, logs, security, cost, EKS intro
Completion evidence	Repeatable environment; Monitored capstone
Section weight	30% of the weighted mock-test average
Assessment	10 MCQ (30 marks) + 4 troubleshooting scenarios (30) + 1 authorised practical lab (40)

Module 7 - Infrastructure as code

Objective: Terraform/CloudFormation, state, review

Observable evidence: Repeatable environment

Concept	Approved knowledge statement	Application behaviour	Evidence
Terraform/CloudFormation	A defined competency within AWS with DevOps. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when terraform/cloudformation is appropriate; apply it in a AWS with DevOps task; verify the result.	Repeatable environment
state	A defined competency within AWS with DevOps. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when state is appropriate; apply it in a AWS with DevOps task; verify the result.	Repeatable environment
review	A defined competency within AWS with DevOps. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when review is appropriate; apply it in a AWS with DevOps task; verify the result.	Repeatable environment

Module 8 - Operations

Objective: CloudWatch, logs, security, cost, EKS intro

Observable evidence: Monitored capstone

Concept	Approved knowledge statement	Application behaviour	Evidence
CloudWatch	A defined competency within AWS with DevOps. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when cloudwatch is appropriate; apply it in a AWS with DevOps task; verify the result.	Monitored capstone
logs	A defined competency within AWS with DevOps. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when logs is appropriate; apply it in a AWS with DevOps task; verify the result.	Monitored capstone
security	A defined competency within AWS with DevOps. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when security is appropriate; apply it in a AWS with DevOps task; verify the result.	Monitored capstone
cost	A defined competency within AWS with DevOps. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when cost is appropriate; apply it in a AWS with DevOps task; verify the result.	Monitored capstone
EKS intro	A defined competency within AWS with DevOps. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when eks intro is appropriate; apply it in a AWS with DevOps task; verify the result.	Monitored capstone

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Repeatable environment; Monitored capstone, return to Modules 7-8, complete one guided practice and then use a new equivalent test form.

NT-C-006 | AWS with DevOps | Skill Ontology & Dependency Map

Atomic skills, learning order and evidence strength

Skill ID	Competency	Depends on	Evidence	Type
NT-C-006-SK01	DevOps foundation	Prerequisite foundation	Version-controlled lab	Critical
NT-C-006-SK02	AWS core	M01 plus earlier foundations	Secure cloud environment	Critical
NT-C-006-SK03	Availability	M02 plus earlier foundations	Highly available design	Supporting
NT-C-006-SK04	Automation	M03 plus earlier foundations	Automation scripts	Supporting
NT-C-006-SK05	Containers	M04 plus earlier foundations	Containerised service	Supporting
NT-C-006-SK06	CI/CD	M05 plus earlier foundations	Working pipeline	Critical
NT-C-006-SK07	Infrastructure as code	M06 plus earlier foundations	Repeatable environment	Supporting
NT-C-006-SK08	Operations	M07 plus earlier foundations	Monitored capstone	Critical

Evidence-strength hierarchy

Strength	Evidence source	Use
High	Trainer-observed practical, viva, working source/project files and reproducible output.	May support verified mastery when rubric threshold is met.
Medium	Scored mock tests, reviewed assignments and documented portfolio artifacts.	Supports mastery with corroboration.
Low	Resume claim, self-report, attendance or video completion without demonstration.	May trigger verification; cannot alone support Apply Now.

CareerTours must not treat attendance, time spent or content opened as skill mastery.

NT-C-006 | AWS with DevOps | Assessment Knowledge & Sample Items

Non-live examples for LLM training and evaluation

Sample ID	Type	Question	Approved answer / rubric anchor
NT-C-006-S01-EX01	Evidence/application	What evidence best demonstrates the combined competency of DevOps foundation and AWS core?	The learner should provide Version-controlled lab; Secure cloud environment and explain how the work applies Linux, networking, Git, delivery lifecycle; IAM, VPC, EC2, S3, RDS. Recall alone is insufficient.
NT-C-006-S01-EX02	Scenario/remediation	A learner reports completing AWS core but cannot produce Secure cloud environment. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-006-S01, request Secure cloud environment, and allow a new equivalent test only after remediation.
NT-C-006-S02-EX01	Evidence/application	What evidence best demonstrates the combined competency of Availability and Automation?	The learner should provide Highly available design; Automation scripts and explain how the work applies Load balancing, Auto Scaling, DNS basics; Shell/Python basics, configuration tasks. Recall alone is insufficient.
NT-C-006-S02-EX02	Scenario/remediation	A learner reports completing Automation but cannot produce Automation scripts. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-006-S02, request Automation scripts, and allow a new equivalent test only after remediation.
NT-C-006-S03-EX01	Evidence/application	What evidence best demonstrates the combined competency of Containers and CI/CD?	The learner should provide Containerised service; Working pipeline and explain how the work applies Docker, images, registries, ECR; Build, test, deploy pipelines; approvals. Recall alone is insufficient.
NT-C-006-S03-EX02	Scenario/remediation	A learner reports completing CI/CD but cannot produce Working pipeline. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-006-S03, request Working pipeline, and allow a new equivalent test only after remediation.
NT-C-006-S04-EX01	Evidence/application	What evidence best demonstrates the combined competency of Infrastructure as code and Operations?	The learner should provide Repeatable environment; Monitored capstone and explain how the work applies Terraform/CloudFormation, state, review; CloudWatch, logs, security, cost, EKS intro. Recall alone is insufficient.
NT-C-006-S04-EX02	Scenario/remediation	A learner reports completing Operations but cannot produce Monitored capstone. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-006-S04, request Monitored capstone, and allow a new equivalent test only after remediation.

Live bank rule: Generate at least three times the live quantity for each section using the approved Infrastructure/Security blueprint. Human-review every scored item before initial production release.

NT-C-006 | AWS with DevOps | Labs, Projects & Scoring Rubrics

Practical proof of learning and ownership

Level	Approved project	Skills evidenced	Required deliverables
Mini 1	Secure web server environment	IAM, VPC, EC2	Brief; working/source file or reproducible environment; output; validation notes; reflection.
Mini 2	Container CI/CD pipeline	Docker, registry, pipeline	Brief; working/source file or reproducible environment; output; validation notes; reflection.
Capstone	Highly available application platform	IaC, deployment, monitoring	Brief; working/source file or reproducible environment; output; validation notes; reflection.

Standard project rubric

Dimension	Weight	What earns credit
Problem framing	15%	Clear objective, user/business need, assumptions and success criteria.
Technical/design accuracy	25%	Correct methods, appropriate tools and sound decisions.
Completeness	15%	Required features, assets, data or workflow are present.
Testing and validation	15%	Important normal, edge, quality or failure cases are checked.
Evidence and reproducibility	15%	Source/working files, setup, inputs, outputs and version information are available.
Documentation	5%	Concise structure, instructions and limitations.
Viva / ownership	10%	Learner can explain decisions, demonstrate work and respond to a change request.

Require least privilege, authorised labs, configuration evidence, logs, monitoring, rollback and documented troubleshooting.

NT-C-006 | AWS with DevOps | Career Roles, Evidence & Gap Actions

Explainable top-three occupation matching

Rank	Occupation	Default weighted skills	Minimum evidence	Gap action
1	Cloud Support Associate	AWS core 34% [critical]; Linux 33% [critical]; networking 33%	Troubleshooting lab portfolio	If a critical skill is below 60 or troubleshooting lab portfolio is missing, return Improve First or Needs Evidence with the linked module/project.
2	Junior DevOps Engineer	Git 25% [critical]; containers 25% [critical]; CI/CD 25%; IaC 25%	Automated delivery project	If a critical skill is below 60 or automated delivery project is missing, return Improve First or Needs Evidence with the linked module/project.
3	AWS Cloud Engineer - Trainee	AWS architecture 34% [critical]; security 33% [critical]; monitoring 33%	Deployed cloud capstone	If a critical skill is below 60 or deployed cloud capstone is missing, return Improve First or Needs Evidence with the linked module/project.

Course readiness	45% section mocks + 20% mini projects + 25% capstone + 10% viva.
Role readiness	45% role-relevant mastery + 25% projects/capstone + 20% aligned mocks + 10% verified resume evidence.
Evidence confidence	Show separately; Apply Now requires at least 70.
Resume extraction	Use only relevant skills, projects, education, experience and portfolio evidence; minimise personal data before external model processing.

A student may be Apply Now for one mapped occupation and Improve First for another. Return independent evidence and gaps for each role.

NT-C-006 | AWS with DevOps | FAQs, Misconceptions & LLM Response Pack

Course-specific retrieval and response patterns

Approved FAQ	Answer
Who is AWS with DevOps for?	Learners targeting cloud support, infrastructure automation and junior DevOps roles.
What should I know before starting?	Basic networking and operating-system knowledge; Linux foundation may be required.
How long is the approved learning journey?	Approved working range: 16-20 weeks / 180-240 guided hours.
What will I be able to demonstrate?	Deploy and operate a secure cloud application using AWS services, containers, CI/CD, infrastructure as code and monitoring.
Does completion guarantee a job?	No. Completion and role readiness are evidence-based guidance; employment, placement, salary and employer selection are not guaranteed.
What if I fail a section?	CareerTours identifies the failed skill tags and module, assigns targeted practice, and permits an equivalent retest after remediation.

Misconception	Approved correction
Attendance proves mastery	False - require tests and observable practical/project evidence.
One total score proves every role fit	False - calculate each occupation using its weighted skills and evidence.
The LLM can add missing details	False - retrieve approved records or state that evidence is insufficient.
Course completion guarantees employment	False - never claim guaranteed job, placement, salary or interview.
AWS with DevOps knowledge never changes	False - version-sensitive tools, laws, taxes, security and platform procedures require review.

Retrieval metadata

```
course_code=NT-C-006; corpus_version=3.0;
record_types=course_profile,module,section,skill,assessment_sample,project,occupation_mapping,faq,misconception;
approval_status=approved; effective_date=2026-08-12
```